

AGNIK DUTTA

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Education

S.R.M. Institute of Science and Technology

B.Tech in Computer Science and Engineering | CGPA: 8.57

Kattankulathur, Chennai

Aug. 2023 – Aug. 2027

Technical Skills

Artificial Intelligence & Machine Learning: TensorFlow, Keras, CNN Architectures, PyTorch, RNN/LSTM, Scikit-learn, OpenCV, NLP, Data Structures & Algorithms

Generative AI & Agentic Systems: LangChain, Agentic Pipeline Design, RAG (Retrieval-Augmented Generation), Vector Databases (ChromaDB), LLM APIs (OpenAI, Groq), Embeddings (TF-IDF, Cosine Similarity), Prompt Engineering

Software Engineering & SDLC: Java, JavaScript, FastAPI, Node.js, React.js, OOP, RESTful APIs, Git, Agile Methodology, Basic System Design, DBMS, Operating Systems, Computer Networks

Data Engineering & Analytics: Python, Pandas, NumPy, PySpark, ETL Pipelines, SQL, Data Modeling, Data Cleaning & Preprocessing, Exploratory Data Analysis, Power BI, Predictive Analytics

Cloud & DevOps: AWS (EC2, S3, Glue, SageMaker), Docker, GitHub Actions (CI/CD Automation), MongoDB, Supabase, REST API Deployment

Experience

Data Science Intern

Jan 2025 – Mar 2025

Zidio | [Certificate](#)

- Built and automated batch-oriented **ETL** pipelines using Python, Pandas, and NumPy to collect, clean, and structure large-scale employee feedback data from multiple sources, applying **data modeling** principles to prepare it for downstream model training and analysis.
- Optimized **SQL** queries (JOIN, GROUP BY, HAVING) across RDBMS tables to evaluate relationships between data fields, then built **Power BI** dashboards and documented **exploratory data analysis** findings to present model-ready insights to technical and non-technical stakeholders.

Business Associate

Jan 2025 – Dec 2025

WellWith / Ayuzera (Shark Tank Startup) | [Link](#)

- Worked directly with cross-functional Data Science and Product stakeholders, translating business goals into actionable **analytics** across 100+ monthly user interactions to improve campaign performance.
- Led Agile teams of 6-10 members in fast-paced, ambiguous environments, owning delivery end-to-end from problem framing to shipped outcomes.

Projects

SkillGap AI – Agentic RAG Recruitment Platform | [GitHub](#)

- Architected an **agentic RAG** pipeline with **FastAPI** and **PyTorch**, orchestrating multi-step retrieval, matching, and **LLM**-based feedback stages over unstructured resume and job description data.
- Engineered a hybrid retrieval system combining **TF-IDF/cosine similarity** with **LangChain**-orchestrated **vector database** (**ChromaDB**) embedding search, improving skill-match accuracy by 20% across 500+ job descriptions.
- Migrated persistence layer to **Supabase** and rebuilt the frontend in **React**, with **GitHub Actions CI/CD** automating deployment and Hoppscotch-driven **quality assurance**.

Neural Style Transfer – Arbitrary Style Synthesis Engine

- Implemented an arbitrary **neural style transfer** system in **PyTorch/Torchvision** using **Adaptive Instance Normalization (AdaIN)** over pre-trained **VGG-19** feature statistics to fuse content and style images.
- Built an interactive **Flask** web application to serve the model, enabling real-time style synthesis from user-uploaded content and style image pairs.

AI Real-Time Gym Coach | [GitHub](#)

- Built a real-time computer vision application using **MediaPipe** pose estimation to track exercise form, integrating a **Groq LLM (Generative AI)** to generate live coaching feedback delivered via gTTS voice output.
- Designed 5 modular exercise-detection models handling real-time video via WebRTC and persisting session data in SQLite, iteratively tuning detection thresholds – covering the full cycle from data capture to inference to output.

Achievements

- Published research paper on "Continuous Context-Aware Skill Gap Detection and Trend-Driven Personalized Learning Path Generator Using **Large Language Models**" at the IEEE - CLICK 2026 international conference. | [Certificate](#)
- Built and pitched solutions across 4 hackathons – Smart India Hackathon (Internal), Hack of Duty, Serve-Smart Hackathon (IIT BHU Jagriti), and The Empathy Encryption Hackathon.
- Selected for Harvard-Affiliated Aspire Leadership Program – admitted as 1 of 500+ global applicants. | [Certificate](#)
- Earned NPTEL certifications in **Data Structures & Algorithms** and **Machine Learning**; solved 140+ problems on LeetCode, and regularly use **ChatGPT** and **GitHub Copilot** to accelerate research and development. | [LeetCode](#)